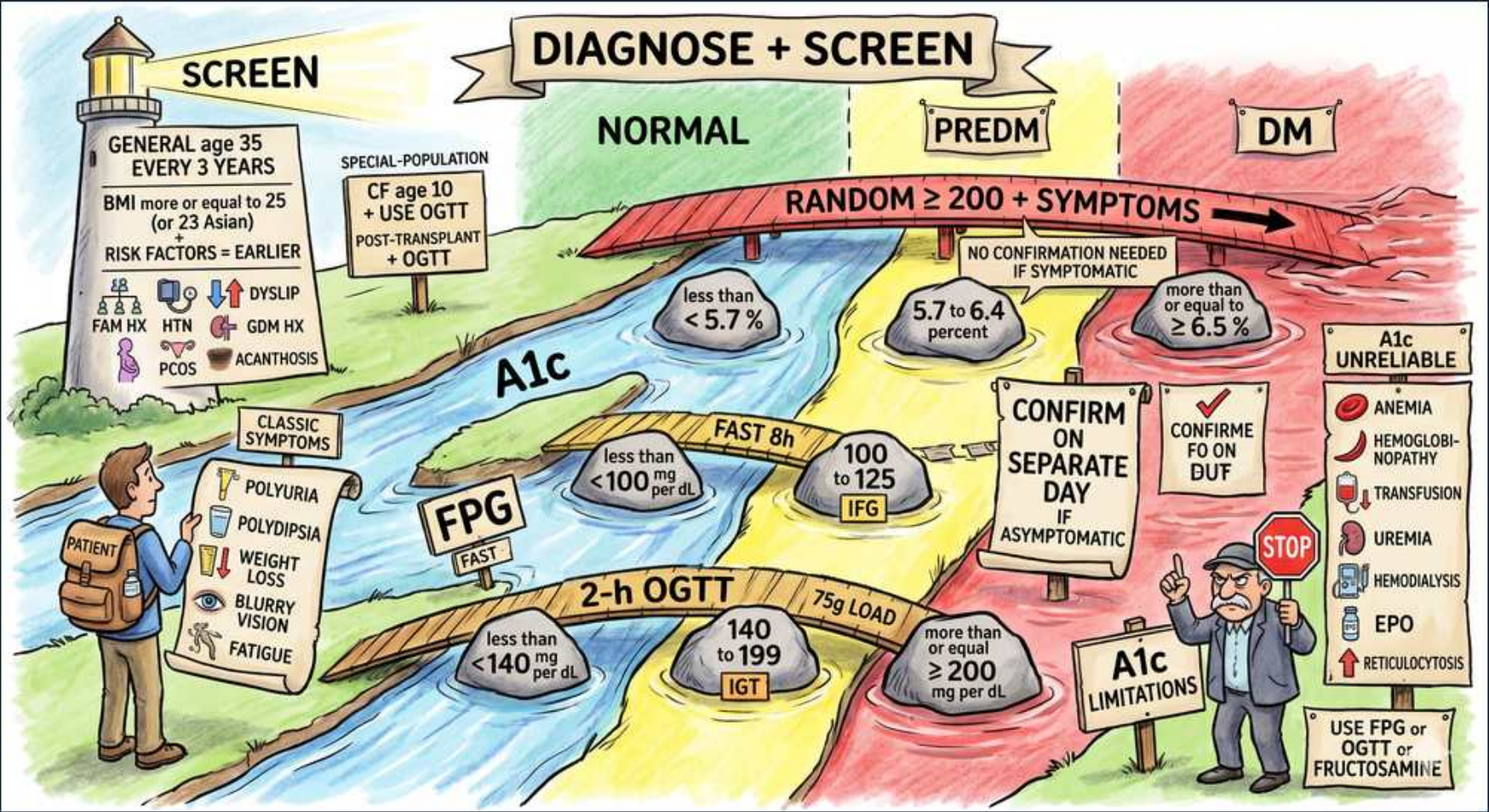


DM — Scene 1 — DM Pathophys & Dx: The Diagnostic Threshold Crossing



1. Lighthouse SCREEN

WHAT YOU SEE

Lighthouse labeled SCREEN, general age 35 every 3 years, BMI ≥ 25 (≥ 23 Asian)

WHAT IT ENCODES

ADA recommends universal screening for type 2 DM starting at age 35, repeated every 3 years if normal. Screen EARLIER (any age) if overweight/obese — BMI ≥ 25 (≥ 23 in Asian Americans, who develop insulin resistance at lower BMI) PLUS one risk factor. Once prediabetes is found, screen annually. Test of choice: A1c, FPG, or 2-h OGTT — any one is acceptable. Mnemonic: '35 every 3.'

BOARD TRAP

WRONG: 'Only screen patients with symptoms (polyuria/polydipsia).' Most T2DM is asymptomatic for years; the whole point of screening is to catch it before symptoms. Also WRONG: the old age-45 cutoff — ADA lowered it to 35 in 2022. Discriminator: screening is age/risk-driven, not symptom-driven.

2. Risk factors banner

WHAT YOU SEE

FAM HX, HTN, DYSLIP, GDM HX, PCOS, acanthosis nigricans listed as RISK FACTORS = EARLIER

WHAT IT ENCODES

Any one risk factor in an overweight patient mandates screening before age 35: first-degree relative with DM, HTN ($\geq 130/80$), dyslipidemia (HDL < 35 or TG > 250), prior gestational DM, PCOS, physical inactivity, CVD, or acanthosis nigricans (a visible sign of severe insulin resistance). Women with prior GDM need lifelong q1-3y screening. High-risk ethnicity (Black, Hispanic, Native American, Asian, Pacific Islander) also counts.

BOARD TRAP

WRONG: 'A patient with PCOS and acanthosis nigricans at age 28 with normal weight does not need screening yet.' Acanthosis and PCOS are insulin-resistance markers — screen regardless of the age-35 rule. Discriminator: risk factors OVERRIDE the age threshold.

3. Special-population sign

WHAT YOU SEE

Sign: CF age 10 + use OGTT, post-transplant + OGTT

WHAT IT ENCODES

Two populations are screened with OGTT, NOT A1c: cystic fibrosis-related diabetes (annual OGTT from age 10) and post-transplant diabetes (NODAT). In CF, A1c is falsely low from altered glycation/RBC turnover; in transplant patients, steroids/calcineurin inhibitors (tacrolimus) cause hyperglycemia that A1c underestimates early. OGTT is the most sensitive test and catches disease A1c/FPG miss.

BOARD TRAP

WRONG: 'Order an A1c to screen the CF patient / post-transplant patient.' A1c is unreliable here and falsely reassuring. Discriminator: when chronicity is short (new transplant) or glycation is altered (CF), the OGTT — not A1c — is correct.

4. RANDOM ≥ 200 + symptoms

WHAT YOU SEE

Red arrow: RANDOM ≥ 200 + SYMPTOMS, no confirmation needed if symptomatic

WHAT IT ENCODES

A random (non-fasting) plasma glucose ≥ 200 mg/dL in a patient with classic hyperglycemic symptoms (polyuria, polydipsia, weight loss) OR in hyperglycemic crisis is diagnostic of DM on a SINGLE test — no confirmatory repeat needed. This is the one scenario where one value seals the diagnosis. All other (asymptomatic) abnormal tests require confirmation.

BOARD TRAP

WRONG: 'Every DM diagnosis must be confirmed on a second day.' False — a symptomatic random glucose ≥ 200 stands alone. Discriminator: symptoms + random ≥ 200 = diagnose now; asymptomatic abnormal value = repeat to confirm.

5. A1c $< 5.7\%$ (Normal)

WHAT YOU SEE

Rock labeled less than $< 5.7\%$ under NORMAL column

WHAT IT ENCODES

A1c $< 5.7\%$ = normal glucose tolerance. A1c reflects average glycemia over ~3 months (lifespan of RBC); it correlates with mean glucose (A1c 6% $\approx$ 126 mg/dL, each 1% $\approx$ 28-29 mg/dL). No fasting required, low day-to-day variability — its main practical advantage.

BOARD TRAP

WRONG: 'A normal A1c rules out diabetes in a patient with anemia.' A1c can be falsely low (normal) with shortened RBC survival. Discriminator: a normal A1c is only trustworthy when RBC turnover is normal.

6. A1c 5.7-6.4% (Prediabetes)

WHAT YOU SEE

Rock 5.7 to 6.4 percent under PREDM

WHAT IT ENCODES

A1c 5.7-6.4% = prediabetes. These patients have ~5-10%/yr conversion to overt DM and elevated CV risk. Management: intensive lifestyle (7% weight loss, 150 min/wk exercise) cuts progression ~58%; metformin is added for highest-risk (BMI ≥ 35 , age < 60 , prior GDM, A1c $\geq 6.0\%$). Prediabetes thresholds: A1c 5.7-6.4, FPG 100-125 (IFG), 2-h OGTT 140-199 (IGT).

BOARD TRAP

WRONG: 'Start a sulfonylurea / start insulin for prediabetes.' Lifestyle is first-line; metformin is the only drug recommended, and only for high-risk subsets. Discriminator: prediabetes = lifestyle +/- metformin, never insulin/SU.

7. A1c $\geq$ 6.5% (DM)

WHAT YOU SEE

Rock more than or equal to $\geq$ 6.5% under DM column

WHAT IT ENCODES

A1c $\geq$ 6.5% diagnoses diabetes. Must be a standardized/NGSP-certified DCCT-aligned assay. If the patient is asymptomatic, confirm with a repeat A1c $\geq$ 6.5 (or a second different test that is also in the diabetic range) on a separate occasion. A1c $\geq$ 6.5 and FPG $\geq$ 126 are independent qualifying criteria — meeting either twice (or two different tests once each) confirms DM.

BOARD TRAP

WRONG: 'A single A1c of 6.6% in an asymptomatic patient diagnoses DM.' One abnormal asymptomatic value is NOT enough — repeat to confirm. Discriminator: asymptomatic requires confirmation; symptomatic random $\geq$ 200 does not.

8. FPG fasting $\geq$ 8h plank

WHAT YOU SEE

FAST plank FPG, less than <100 normal, 100-125 IFG

WHAT IT ENCODES

Fasting plasma glucose requires $\geq$ 8 hours no caloric intake. Cutoffs: <100 normal, 100-125 = impaired fasting glucose (prediabetes), $\geq$ 126 mg/dL = diabetes. FPG mainly reflects hepatic glucose output/basal insulin. Cheap and widely used but has more day-to-day variability than A1c and requires fasting.

BOARD TRAP

WRONG: 'A non-fasting glucose of 130 diagnoses DM by the FPG criterion.' The 126 cutoff applies ONLY to a true $\geq$ 8h fasting sample. Discriminator: a random 130 is not diagnostic (need $\geq$ 200 + symptoms, or a fasting $\geq$ 126).

9. 2-h OGTT 75g load

WHAT YOU SEE

2-h OGTT plank, <140 normal, 140-199 IGT, $\geq$ 200 DM, 75g LOAD

WHAT IT ENCODES

75g oral glucose tolerance test, glucose measured at 2 hours: <140 normal, 140-199 = impaired glucose tolerance (prediabetes), $\geq$ 200 mg/dL = diabetes. Most SENSITIVE of the three tests — catches early DM that FPG/A1c miss (post-prandial defect). Standard for gestational DM screening and for CF/post-transplant. Note GDM uses different (lower) thresholds.

BOARD TRAP

WRONG: 'A1c is the most sensitive test for early diabetes.' The 2-h OGTT detects the earliest (post-load) abnormalities and is the most sensitive. Discriminator: OGTT > FPG > A1c for sensitivity; A1c wins on convenience, not sensitivity.

10. Confirm on separate day

WHAT YOU SEE

Sign CONFIRM ON SEPARATE DAY IF ASYMPTOMATIC

WHAT IT ENCODES

In an ASYMPTOMATIC patient, a single abnormal test must be confirmed by repeating the same test, or by a second different test, that again falls in the diabetic range — ideally a new sample. Exception: if two DIFFERENT tests from the SAME draw are both diagnostic, that confirms DM without a second visit. If two tests are discordant, repeat the one that was abnormal.

BOARD TRAP

WRONG: 'Diagnose DM from one elevated FPG in an asymptomatic screening patient.' Confirmation is required. Discriminator: asymptomatic = confirm; symptomatic crisis/random $\geq$ 200 = no confirmation.

11. A1c UNRELIABLE list

WHAT YOU SEE

Sign A1c UNRELIABLE: anemia, hemoglobinopathy, transfusion, uremia, hemodialysis, EPO, reticulocytosis

WHAT IT ENCODES

A1c is unreliable whenever RBC lifespan or hemoglobin glycation is abnormal. FALSELY LOW: anything shortening RBC survival or increasing young cells — hemolytic anemia, acute blood loss, recent transfusion, EPO therapy, reticulocytosis, pregnancy, ESRD/hemodialysis. FALSELY HIGH: iron/B12 deficiency anemia (older RBCs), splenectomy, alcoholism. Hemoglobinopathies (HbS, HbC) interfere with the assay either way.

BOARD TRAP

WRONG: 'Use A1c to manage the dialysis patient / patient on EPO / patient with sickle cell.' These falsify A1c (usually falsely low in high-turnover states). Discriminator: altered RBC turnover = abandon A1c, use FPG/OGTT/fructosamine.

12. STOP — A1c limitations

WHAT YOU SEE

STOP sign, A1c LIMITATIONS

WHAT IT ENCODES

STOP: when any condition altering RBC turnover or hemoglobin is present, do NOT use A1c to diagnose OR to monitor — it will mislead in both directions. Recognize the falsely-normal A1c trap: a hemolytic-anemia patient with true hyperglycemia can show a deceptively normal A1c. Switch to glucose-based testing immediately.

BOARD TRAP

WRONG: 'Trust a normal A1c in a patient with a known hemoglobinopathy.' The assay is invalid; a normal value does not exclude DM. Discriminator: in any A1c-unreliable state, the absence of an elevated A1c proves nothing.

13. Use FPG/OGTT/fructosamine

WHAT YOU SEE

Sign USE FPG or OGTT or FRUCTOSAMINE

WHAT IT ENCODES

When A1c is invalid, diagnose/monitor with FPG, OGTT, or fructosamine. Fructosamine reflects glycated serum proteins (mainly albumin) over the prior ~2-3 weeks — independent of RBC turnover, so it is the go-to in hemolysis, recent transfusion, or pregnancy. (Caveat: fructosamine is itself falsely low in hypoalbuminemia/protein-losing states.) For diagnosis specifically, OGTT/FPG are the validated tools.

BOARD TRAP

WRONG: 'Order another A1c in 3 months to monitor a dialysis patient.' Repeat A1c stays unreliable. Discriminator: substitute fructosamine (or glucose monitoring), not a repeat A1c.

14. Classic symptoms patient

WHAT YOU SEE

Patient with polyuria, polydipsia, weight loss, blurry vision, fatigue

WHAT IT ENCODES

Classic hyperglycemia symptoms = the 3 P's plus more: polyuria and nocturia (osmotic diuresis once glucose exceeds the renal threshold ~180 mg/dL → glucosuria), polydipsia (compensating water loss), unexplained weight loss (catabolism/glycosuric calorie loss — prominent in T1DM/DKA), blurry vision (osmotic lens swelling), fatigue, and recurrent infections (candidiasis). Symptoms + random glucose ≥ 200 = diagnostic without confirmation.

BOARD TRAP

WRONG: 'Weight gain is a classic presenting symptom of new hyperglycemia.' New marked hyperglycemia causes weight LOSS (calorie loss via glucosuria + catabolism). Discriminator: weight loss with polyuria/polydipsia points to new DM (especially T1DM), not weight gain.
